

Assignment No-03

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Course : CSE320

Section : 20

Ans to the Q No-1

TDM or Time Division Multiplexing is a technique used in telecommunication to transmit multiple signal simultaneously over a single communication channel by allocating each signal a specific time slot. We can divide TDM into two different schemes: synchronous and statistical TDM. The main ~~reason~~ difference between synchronous and statistical TDM is in between time slots. An output slot in synchronous TDM is fully occupied with data unit. On the other hand in statistical TDM a slot needs to carry data as well as the address of the destination.

Synchronous TDM:

Advantages:

- (i) Ensures ~~efficient~~ use of available bandwidth.
- (ii) Data rate of the link is n times faster and the unit duration is n times shorter.

⑦ No need of addressing because synchronization pre-assigned relationships between input and output serves as an address.

⑧ The system works according to time slots so error identification might be easy.

Disadvantages

① Empty slots if there is no data from sender.

② There can be inefficiency in bandwidth if input lines have no data to send. It will result in waste of bandwidth.

~~③ Extra slots for idle~~

Statistical TDM:

Advantages:

① Statistical TDM dynamically allocates time slots based on actual data traffic allowing for more efficient use of bandwidth compared to synchronous TDM.

(2)
II There is an address with each data unit so there will be no empty slots.

~~III A slot carries both data and an address~~

III It doesn't require synchronization bits.

Disadvantages

i) Dynamically allocation may cause bandwidth efficiency but it takes more time to manage time slots.

ii) As multiple channels share same medium so packet loss ~~and~~ may occur.

iii) The process is complex.

Ans to the Q No-2

Time Division Multiplexing (TDM), Frequency Division Multiplexing (FDM), and Wavelength Division Multiplexing (WDM) are all multiplexing techniques used to transmit multiple signals simultaneously over single communication channel. The characteristics of TDM that differs from WDM and FDM is -

① Allocation: In FDM the channel is divided into fixed time slots and each signal is assigned its own time slot. In FDM the available bandwidth is divided into multiple frequency bands and each signal is assigned a certain frequency band. In WDM multiple signals are transmitted simultaneously over same medium but each signal is assigned a unique wavelength.

③

ii) Medium of Transmission: TDM is used in both wired and wireless communication system. FDM is mostly used in wired communication specially in telephone lines. It can also be used on wireless, Such as radio broadcasting, cable television. WDM is used in optical fibre communication system where multiple signals are transmitted over same optical ~~fiber~~ fiber using different wavelength.

Ans to the Q No-3

Interleaving: Interleaving is a technique used in TDM to enhance the efficiency of data transmission. It can be visualized as two fast rotating switches one on the multiplexing side and the other on the demultiplexing side. The switches are synchronized and rotate at the same speed but in opposite direction.

On the multiplexing side as the switch opens in favour of a connector, ~~that~~ that connection ~~op~~ has the opportunity to send a unit onto the path. On the demultiplexing side, as the switch opens in favour of the connection, the connection has the opportunity to receive a unit from the path. This process is called interleaving.

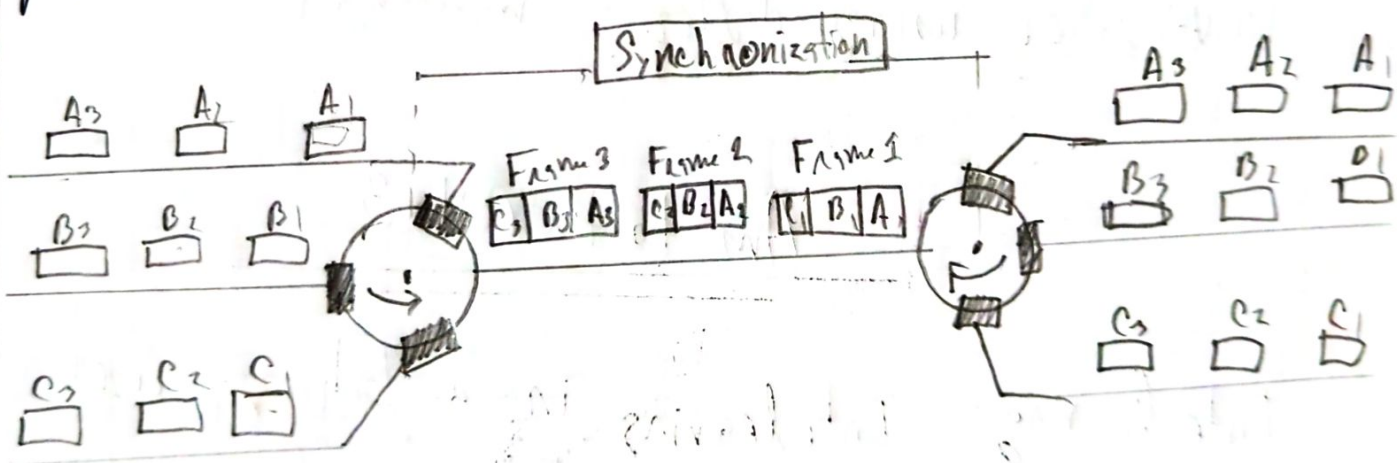


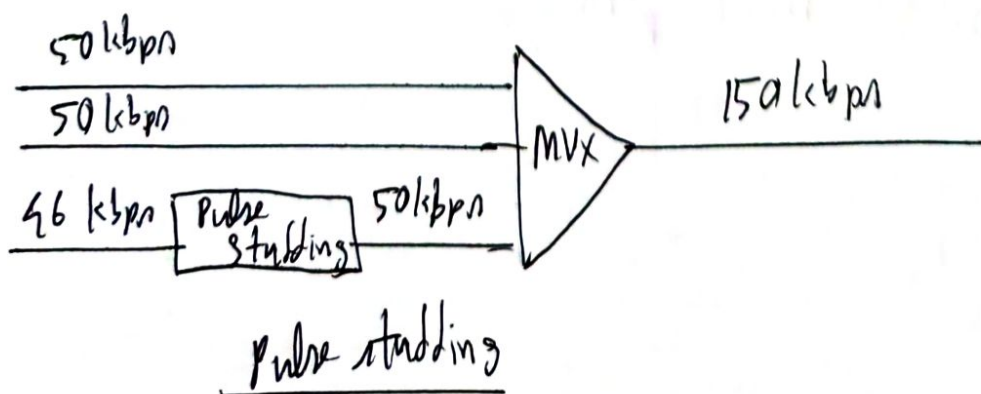
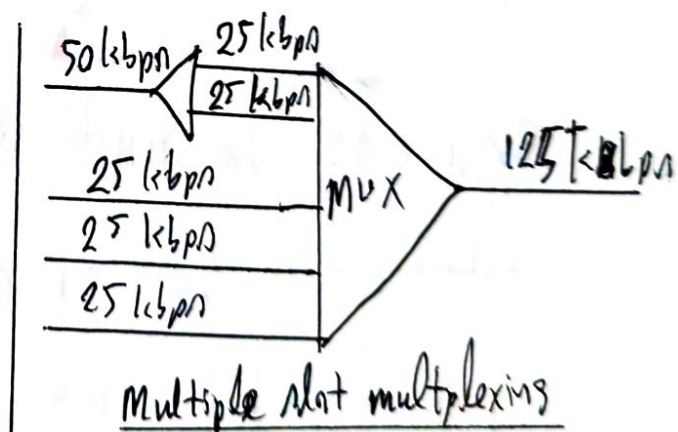
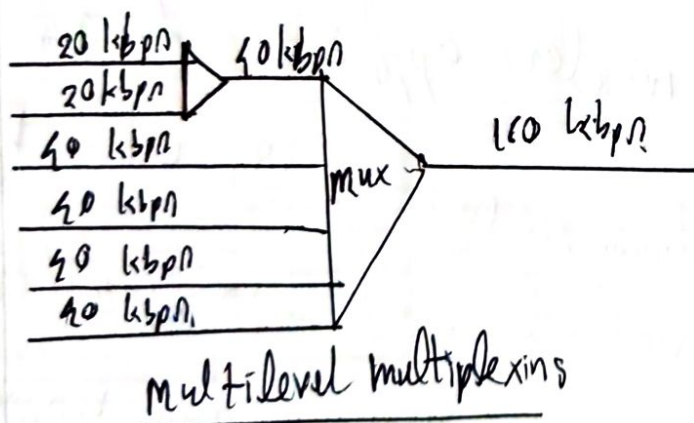
Figure: Interleaving

Data rate management techniques used in FDM:

① Multi-level multiplexing: A technique used when data rate of an input line is a multiple of others.

① Multiple Slot Allocation: More efficient to allot more than one slot in a frame to a single input line.

② Pulse ^{Stuffing} ~~Stuffing~~: Sometimes the bit rates of sources are not multiple integers of each other. ~~Therefore~~ The solution is to make the highest input data rate the dominant data rate and then add dummy bits to the input lines with slower rates. This will increase their rates. This technique is called pulse stuffing.



Ans to the Q No-4

To obtain bandwidth efficiency the available bandwidth of a link is divided into the sources. Combining signals from different sources to fit into a larger bandwidth and then expanding the original bandwidth. This must be done using a process that is independent of the original signal. This process is called spread spectrum.

Spreading is used in wireless applications to avoid intruders or there is extra security. So while FDM, TDM and WDM are available, spreading spectrum provides more privacy and allows redundancy.

Ans to the Q No-5

Frequency division multiplexing (FDM) and Frequency Hopping Spread Spectrum (FHSS) are both techniques used to transmit multiple signals over a shared channel.

Difference between FDM and FHSS is -

① Basic Principle:

FDM: In FDM the available bandwidth is divided into multiple non-overlapping frequency bands, where each signal is assigned to a certain frequency band for transmission.

FHSS: FHSS is a technique that uses M different carrier frequencies that are modulated by same signal. At one moment the signal modulates one carrier frequency and at the next the signal modulates another carrier frequency which is called hopping.

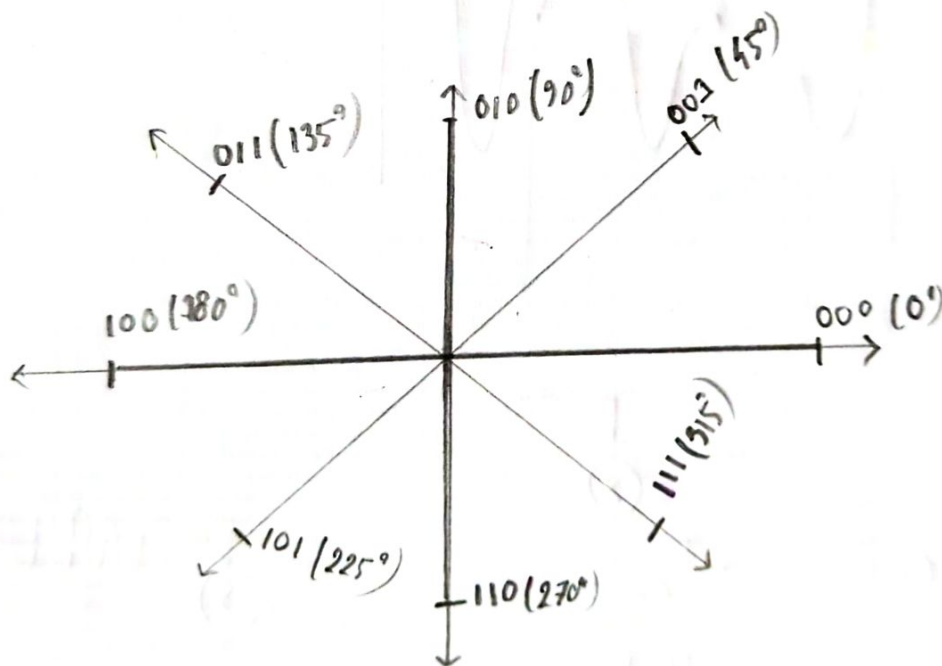
ii) Utilization of Bandwidth:

• FDM: FDM divides the available bandwidth into fixed frequency bands. ~~where~~ So the bandwidth is statically allocated. If there are M stations each station will use $1/M$ bandwidth but the allocation is fixed.

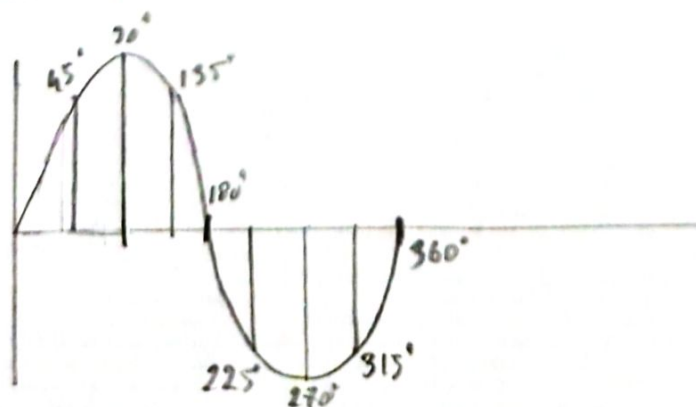
• FHSS: If the number of hopping frequencies is M we can multiplex M channels in one by using the same bandwidth. Each hop $M-1$ frequencies can be used. So in FHSS each station uses $1/M$ of the bandwidth but the allocation changes hop to hop.

Ans to the Q No-6

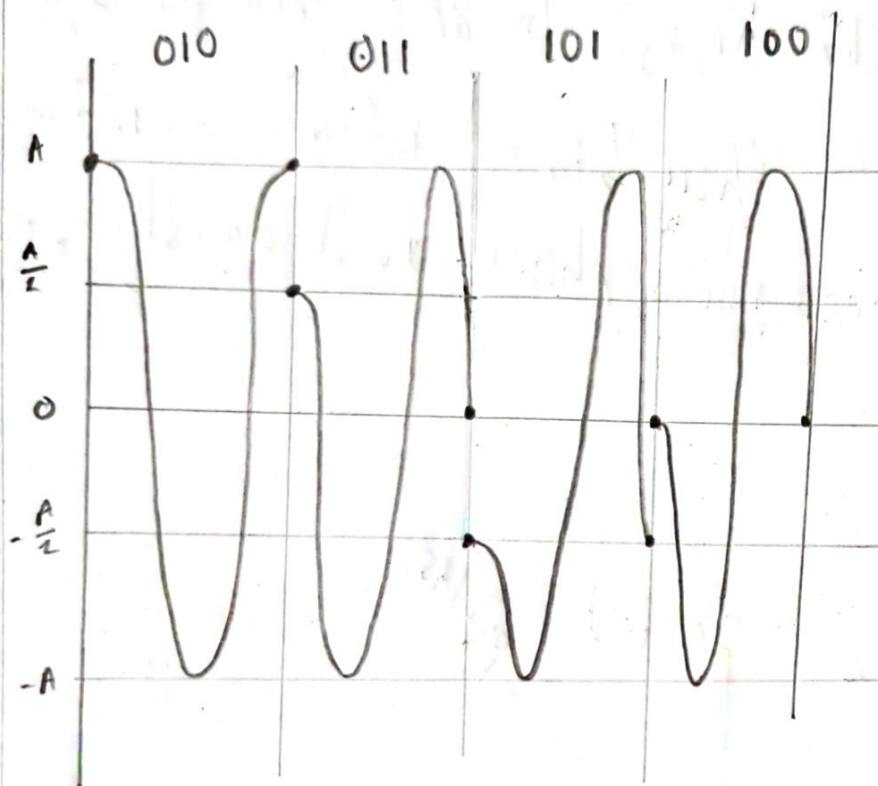
If we had to send 3 bit per signal element then we have to use 8PSK (8 Phase Shift Keying). In 8PSK there are eight possible phases. Therefore three bits of information within each time slot. The corresponding phases will equally separated by 45° degree.



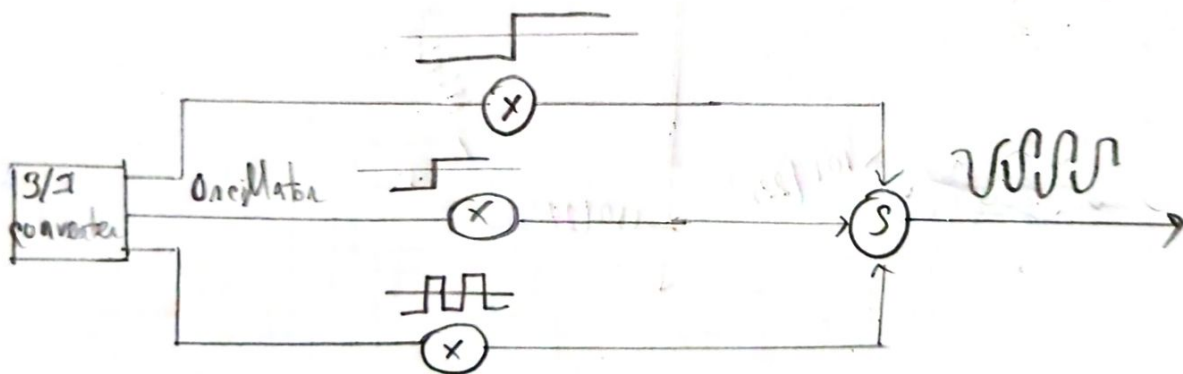
So the amplitudes will be -



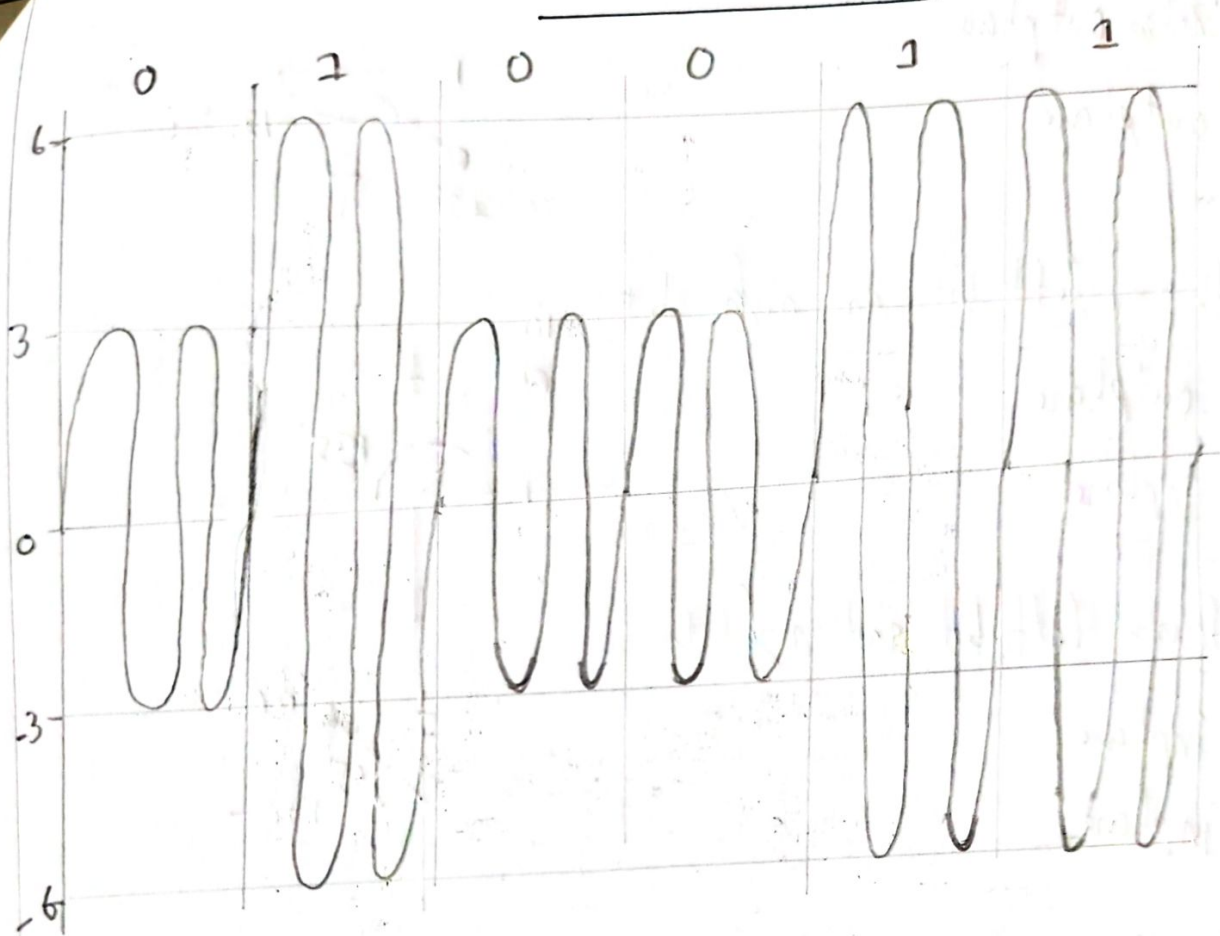
From figure-5.11 if we substitute the values from 2 bit to 3 bits,



Now)



Ans to the Q No-7



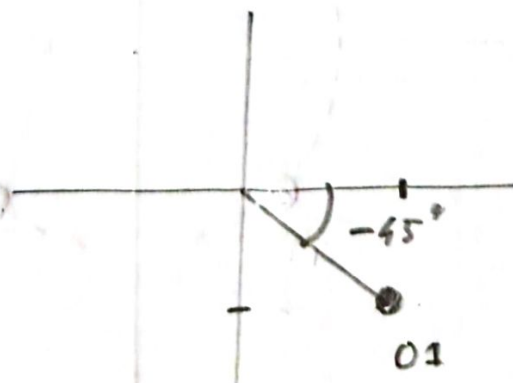
Ans to the Q No-8

Let's draw the diagram part by part.

For the first left bit and right bit.

0 $\rightarrow$ inphase.

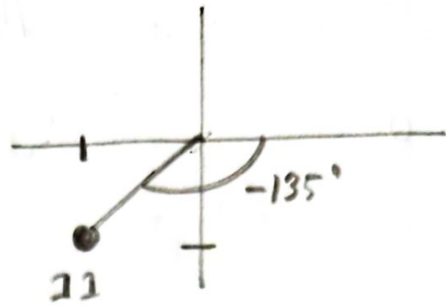
1 $\rightarrow$ outphase.



For second left bit and right bit

1 $\rightarrow$ ~~out~~ outphase

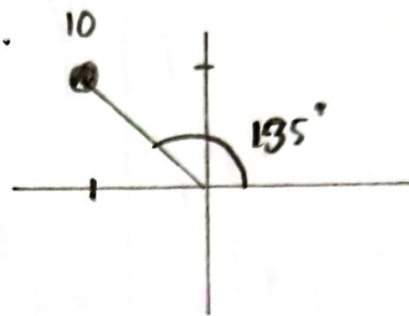
1 $\rightarrow$ outphase.



For third left bit and right bit

1 $\rightarrow$ outphase

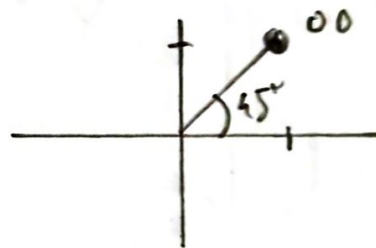
0 $\rightarrow$ inphase



For fourth left bit and right bit

0 $\rightarrow$ inphase

0 $\rightarrow$ inphase.



Constellation diagram for the QPSK is —

